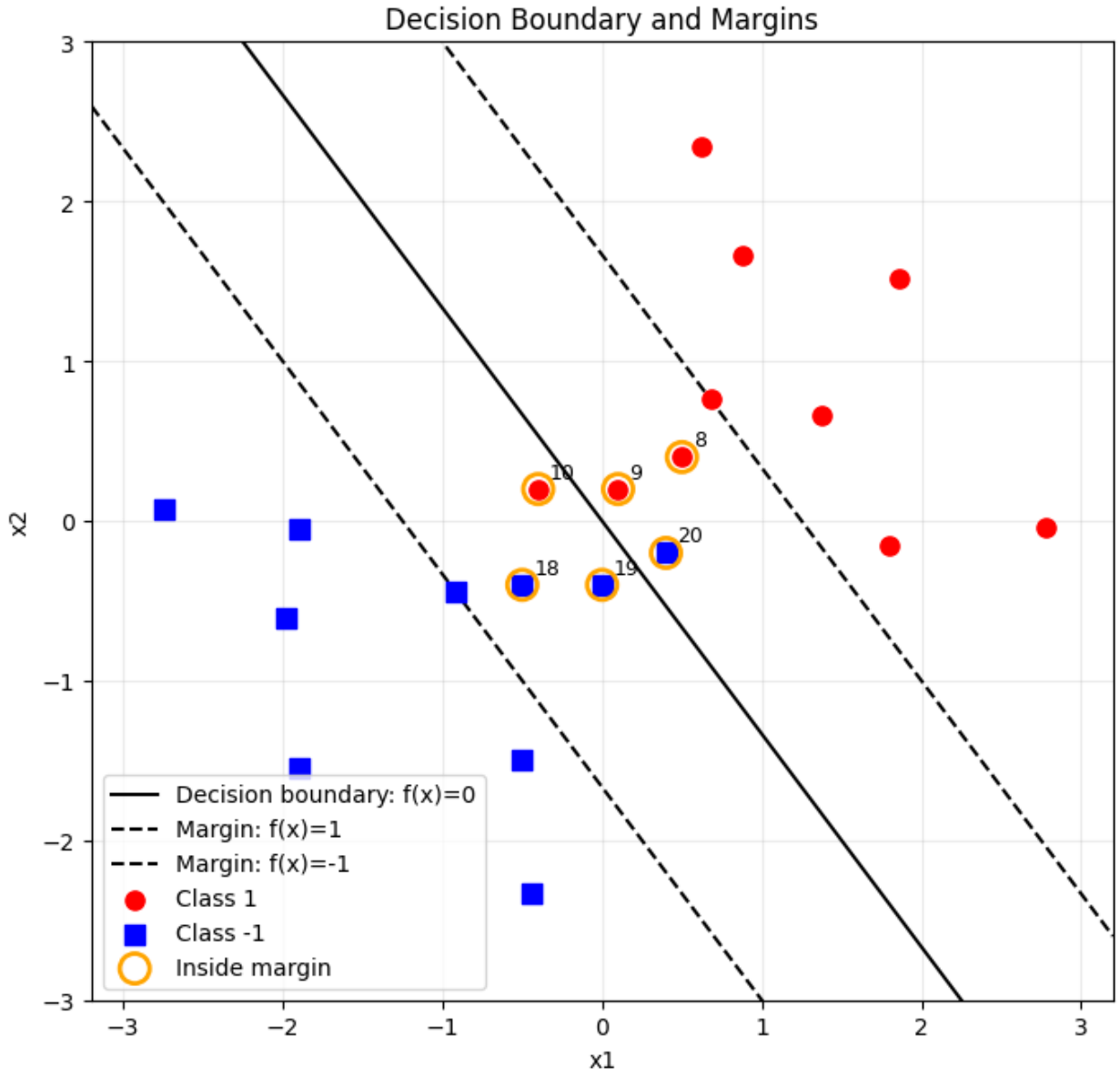
Question 1: Boundary and margin

a. Calculate the distance from every observation to the decision boundary and its functional margin $yf(x)$. In a table, report these values and whether each observation is inside, on, or beyond the margin for its class. Compare $yf(x)$with 1 to make this distinction.

Question 1: Distance and Margin

observation	x1	x2	y	f(x)	Distance	Functional margin	Position	Classification
1	0.68	0.76	1	1.00	1.00	1.00	On	Correct
2	1.80	−0.15	1	1.35	1.35	1.35	Beyond	Correct
3	1.38	0.66	1	1.50	1.50	1.50	Beyond	Correct
4	0.88	1.66	1	1.70	1.70	1.70	Beyond	Correct
5	0.62	2.34	1	1.90	1.90	1.90	Beyond	Correct
6	2.78	−0.04	1	2.20	2.20	2.20	Beyond	Correct
7	1.86	1.52	1	2.40	2.40	2.40	Beyond	Correct
8	0.50	0.40	1	0.64	0.64	0.64	Inside	Correct
9	0.10	0.20	1	0.20	0.20	0.20	Inside	Correct
10	−0.40	0.20	1	−0.20	0.20	−0.20	Inside	Misclassified
11	−0.92	−0.44	−1	−1.00	1.00	1.00	On	Correct
12	−0.50	−1.50	−1	−1.30	1.30	1.30	Beyond	Correct
13	−1.90	−0.05	−1	−1.55	1.55	1.55	Beyond	Correct
14	−0.44	−2.33	−1	−1.75	1.75	1.75	Beyond	Correct
15	−1.98	−0.61	−1	−1.95	1.95	1.95	Beyond	Correct
16	−2.74	0.07	−1	−2.15	2.15	2.15	Beyond	Correct
17	−1.90	−1.55	−1	−2.45	2.45	2.45	Beyond	Correct
18	−0.50	−0.40	−1	−0.64	0.64	0.64	Inside	Correct
19	0.00	−0.40	−1	−0.24	0.24	0.24	Inside	Correct
20	0.40	−0.20	−1	0.20	0.20	−0.20	Inside	Misclassified

b. Plot the observations, the boundary $f(x) = 0$, and the margin lines $f(x) = \pm 1$. Distinguish the two classes and identify the observations inside the margin in your plot.



c. Report how many observations are inside, on, and beyond the margin. Which observations are misclassified by this decision rule?

```
Question 3: Margin Summary
Position  Count                          Observation IDs
  Inside      6                    8, 9, 10, 18, 19, 20
      On      2                                  1, 11
  Beyond     12 2, 3, 4, 5, 6, 7, 12, 13, 14, 15, 16, 17
```

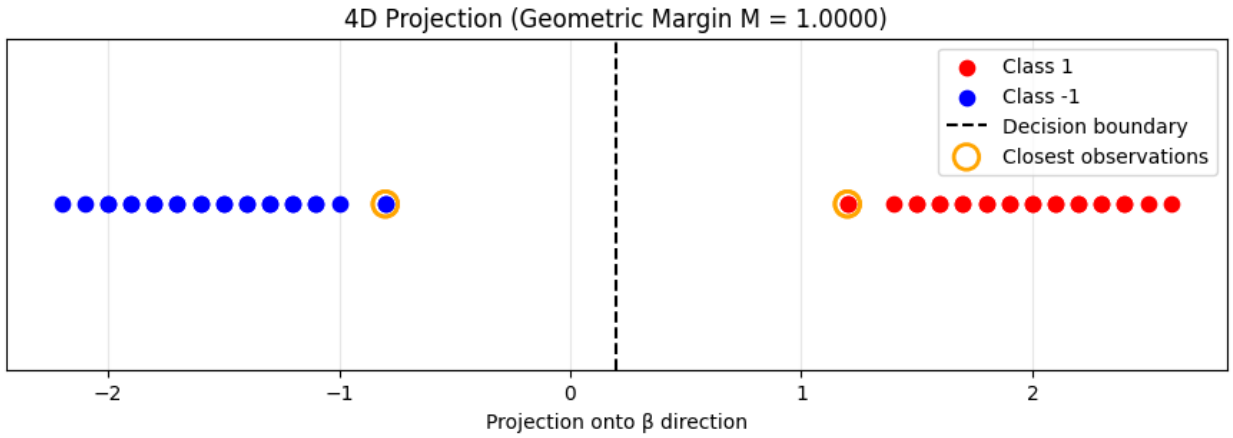
```
Misclassified observations:
Count: 2
Observation IDs: 10, 20
```

Question 2: Margin in four dimensions

a. Project the observations onto the direction of $\boldsymbol{\beta}$and plot their positions on one axis. Distinguish the two classes and mark the decision boundary. Use the plot to calculate the geometric margin M, the distance from the boundary to the closest observation.

```
Geometric margin M = 1.0
```

```
Closest observations:
 observation    x1    x2    x3    x4  y  f(x)  Projection  Distance
           4 -1.26 -0.27 -1.98  0.96  1   5.0         1.2       1.0
          15 -6.16  1.18  0.32  0.11 -1  -5.0        -0.8       1.0
          33  7.74  1.23  1.02 -0.54  1   5.0         1.2       1.0
          36  2.84  0.43 -0.43 -2.14 -1  -5.0        -0.8       1.0
```



b. Express the same decision boundary using the SVM normalization in which the smallest functional margin $y_i f(\boldsymbol{x}_i)$is set to 1. What are the corresponding values of $\boldsymbol{\beta}$and β_0?

```
Minimum original functional margin: 5.0
```

```
Normalized beta: [ 0.2  0.4 -0.4  0.8]
```

```
Normalized beta0: -0.2
```

```
Minimum new functional margin: 0.9999999999999998
```

Question 3: Peer review of a Homework 7 skill

https://github.com/NeelN86/stat432-fall2026/issues/4

Skill: It requires QDA even when a training class has few observations for its covariance estimate. That can make the prescribed analysis fail; specify when to skip QDA and how to report why.

Report — figure: It plots PCA dimension and blend weight on the same x-axis. That makes the curves harder to compare; use separate panels.

Report — parameter counts: The table's PCA-model parameter counts omit the fitted PCA transformation, but the skill uses parameter count to define "simplest" in its one-SE rule. This can affect model selection; define a consistent counting rule and apply it to both selection and the table.